

OC Core 1.3 / Biology Domain Packet

Status Snapshot

- Current state: EMPIRICAL_HARD_CLOSED
- Target state: EMPIRICAL_HARD_CLOSED
- Quantitative pass result: PASS
- Next required action: MAINTAIN_REPLAY_DISCIPLINE
- Replay status: PASS_REPLAYABLE

Theorem-to-Observable Map

- K4/K5/K6/K7 biological organization claims are restricted to bounded state-transition and response-signature lanes.
- The packet currently targets measurable expression and cell-state trajectories rather than unrestricted biological universalization.
- Open biological assay repositories are the first-line evidence route; if insufficient, the program escalates to institute-run measurement.

Benchmark Dataset Manifest

- BIO_BENCH_001: expression_signature_residual | NCBI GEO | <https://www.ncbi.nlm.nih.gov/geo/> | held-out: reserve one time-series family for held-out replay
- BIO_BENCH_002: state_transition_order_error | NCBI GEO | <https://www.ncbi.nlm.nih.gov/geo/> | held-out: reserve one cell-state transition family
- BIO_BENCH_003: response_onset_residual | official open biological assay repositories | <https://www.ncbi.nlm.nih.gov/geo/> | held-out: reserve one response-signature family

Pinned Official Route Protocol

- BIO_ROUTE_001: NCBI GEO (OFFICIAL_OPEN_DATA_HUB) -> <https://www.ncbi.nlm.nih.gov/geo/> | purpose: Open assay and expression benchmark families

Measurable Outputs

- expression_signature_residual
- state_transition_order_error
- response_onset_residual

Measurement Schema

Choose one bounded biological lane and evaluate predicted state transitions or response signatures against open assay data.

Acceptance Criterion

The promoted biological packet reproduces the declared measurable signatures on held-out datasets without widening the claim boundary.

Falsifier

If the bounded signature family fails on held-out datasets, the biological packet remains frontier-bound.

Quantitative Acceptance Thresholds

- coverage_ratio_required: 1.0
- expected_calibration_error_max: 0.05
- normalized_error_max_max_sigma: 5.0
- normalized_error_mean_abs_max_sigma: 1.0
- normalized_error_p95_max_sigma: 2.5
- tail_breach_count_max: 0

Replay Harness

- Replay command: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py --domain-id BIOLOGY`
- Replay program status: `ACTIVE_REPLAY_LANE`
- Execution protocol id: `OC13::EXECUTION::BIOLOGY::STATE_TRANSITIONS_AND_RESPONSE_SIGNATURES`
- Evidence bar: `HYBRID_ESCALATION`

Institute-run Escalation

- Current escalation status: `NOT_REQUIRED`
- Escalation trigger: `OPEN_DATA_COVERAGE_LT_1_0_OR_HELD_OUT_CASES_LT_30_OR_SIGNATURE_CLASS_UNDERCONSTRAINED`
- Measurement wave id: `INSTITUTE_RUN::BIOLOGY::WAVE_3A`
- Measurement plan: Run a bounded institute observation or assay campaign only for the explicitly declared biological lane; no widening of biological scope is allowed.

Blocking IDs

- None.